

OS SKILL 9

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

char previous_dir[PATH_MAX] = "";

void builtin_pwd() {
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("Current directory: %s\n", cwd);
    else
        perror("getcwd");
}

void builtin_cd(const char *path) {
    char current[PATH_MAX];

    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return;
    }

    if (strcmp(path, "-") == 0) {
        if (strlen(previous_dir) == 0) {
            printf("No previous directory available\n");
            return;
        }

        char temp[PATH_MAX];
        strcpy(temp, previous_dir);

        if (chdir(temp) == 0) {
            strcpy(previous_dir, current);
            printf("Changed to previous directory\n");
            builtin_pwd();
        } else {
            perror("cd");
        }

        return;
    }
}
```

```
    if (chdir(path) == 0) {
        strcpy(previous_dir, current);
        printf("Directory changed successfully\n");
        builtin_pwd();
    } else {
        perror("cd");
    }
}

void builtin_echo(const char *text) {
    printf("%s\n", text);
}

void dispatch(const char *command, const char *argument) {
    if (strcmp(command, "pwd") == 0) {
        builtin_pwd();
    }
    else if (strcmp(command, "cd") == 0) {
        builtin_cd(argument);
    }
    else if (strcmp(command, "echo") == 0) {
        builtin_echo(argument);
    }
    else {
        printf("Invalid command: %s\n", command);
    }
}
```

```

}

int main() {
    printf("=== Initial Working Directory ===\n");
    dispatch("pwd", "");

    printf("\n=== Change Directory Test ===\n");
    dispatch("cd", "/tmp");

    printf("\n=== Updated Working Directory ===\n");
    dispatch("pwd", "");

    printf("\n=== Invalid Path Test ===\n");
    dispatch("cd", "/wrong/path");

    printf("\n=== Previous Directory Test ===\n");
    dispatch("cd", "-");

    printf("\n=== Built-in Echo Test ===\n");
    dispatch("echo", "Hello from built-in command");

    printf("\n=== Invalid Command Test ===\n");
    dispatch("wrongcommand", "");

    printf("\n=== Navigation and Dispatch Test Complete ===\n");

    return 0;
}

```

```

nishanth@LAPTOP-6Q3ON07H: ~/task9
nishanth@LAPTOP-6Q3ON07H:~$ mkdir task9
nishanth@LAPTOP-6Q3ON07H:~$ cd task9
nishanth@LAPTOP-6Q3ON07H:~/task9$ nano task9.c
nishanth@LAPTOP-6Q3ON07H:~/task9$ gcc task9.c -o task9
nishanth@LAPTOP-6Q3ON07H:~/task9$ ./task9

=== Initial Working Directory ===
Current directory: /home/nishanth/task9

=== Change Directory Test ===
Directory changed successfully
Current directory: /tmp

=== Updated Working Directory ===
Current directory: /tmp

=== Invalid Path Test ===
cd: No such file or directory

=== Previous Directory Test ===
Changed to previous directory
Current directory: /home/nishanth/task9

=== Built-in Echo Test ===
Hello from built-in command

=== Invalid Command Test ===
Invalid command: wrongcommand

=== Navigation and Dispatch Test Complete ===
nishanth@LAPTOP-6Q3ON07H:~/task9$

```